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(1) Research Program/Project/Study Title: Performance of Different Cacao (<i>Theobroma cacao</i> L) Varieties, Intercrop with Dwarf Coconut Varieties Under Capiz Condition	
Proponent Name:	Dr. Salvacion J. Legaspi
Academic Rank:	Prof. VI
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Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	Region VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity	(8) Discipline/Field of Specialization

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Coconut (*Cocos nucifera*) is an important crop in tropical regions, known for its versatility and economic significance. Cacao (*Theobroma cacao*), commonly known as chocolate tree or cocoa tree is another important crop in tropical regions due to its high demand for chocolate production worldwide (FAOSTAT 2021). Both coconut trees (palms) as well as cacao trees are widely grown in tropical regions including the Philippines.

Farmers utilize a variety of techniques to maximize the use of their land, time, and resources while growing economically important crops. We must recognize that when a portion of a farmer's farm is not used successfully or no crops are grown at a specific period, the farmer suffers a loss. As a result, a cropping plan is frequently implemented by combining several species or kinds of commercial crops; one of these cropping patterns or schemes is the intercropping system (Gregorio, 2016).

Intercropping is a farming practice that involves growing two or more crops simultaneously on the same piece of land. This technique has gained popularity in recent years due to its potential benefits, such as increased productivity, reduced production costs, and improved soil health, as well as its ability to mitigate risks associated with monoculture farming. Intercropping can lead to increased productivity due to the complementary nature of the crops being grown together. According to a study conducted in Papua New Guinea, coconut intercropped with cacao resulted in a 35% increase in coconut yield and a 25% increase in cacao yield compared to monoculture (Kamanga et al., 2019).

This study will seek to utilize dwarf coconut varieties in the CBTU. It also intends to make use of the open areas between mature coconut trees. This also aims to give useful information about the productivity of coconut-based agricultural systems intercropped with other crop commodities.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The objective of this research study is to evaluate the performance of different cacao varieties intercropped with dwarf coconut varieties under Capiz condition. Specifically, the study aims to:

1. Identify the growth and production performance of cacao dwarf coconut varieties when intercropped under Capiz condition.

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2. Determine the optimal planting density and spacing for both crops to maximize yield and minimize competition for resources.
3. Evaluate the impact of intercropping on the quality of cacao and coconut products.
4. Assess the economic feasibility of intercropping of cacao and coconut.
5. Provide recommendations for farmers in Capiz and other similar regions on the best intercropping practices for cacao and dwarf coconut varieties.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Maximizing land usage through appropriate intercropping technologies is critical to sustainable agriculture. The Coconut-Cacao Farming System is an excellent example of a successful intercropping model. According to the study done by Adam et al, Cocoa and Coconut Intercrop Farming reveals good intercrop, as assessed by the distance between coconut and cocoa, which is generally 3.5m on average.

Yield:

Several studies have reported higher yields of cacao when intercropped with coconut compared to monoculture . In a study conducted in Papua New Guinea, the intercropping of cacao with coconut resulted in a 30% increase in cacao yield compared to monoculture (Kenny et al., 2006). In another study in the Philippines, the intercropping of cacao with dwarf coconut resulted in a 26% increase in cacao yield compared to monoculture (Garcia et al., 2011). The higher yields in intercropping systems may be attributed to the shade provided by the coconut trees, which reduces the temperature and humidity, and enhances soil moisture retention, leading to better growth and development of cacao.

Quality:

The intercropping of cacao with coconut has also been found to have a positive impact on the quality of cacao beans. In a study conducted in Indonesia, the intercropping of cacao with coconut resulted in higher bean size and lower moisture content compared to monoculture (Sutisna et al., 2017). In another study in the Philippines, the intercropping of cacao with dwarf coconut resulted in higher bean size and lower moisture content compared to monoculture (Garcia et al., 2011). The higher bean size and lower moisture content may be attributed to the improved microclimate provided by the coconut trees, which reduces the risk of mold and insect damage and enhances the maturation of cacao beans.

Economic feasibility:

The economic feasibility of cacao intercropped in coconut depends on various factors, such as market demand, production costs, and profitability. In a study conducted in Papua New Guinea, the intercropping of cacao with coconut resulted in a higher net income compared to monoculture, due to the higher yields and prices of both crops (Kenny et al., 2006). In another study in the Philippines, the intercropping of cacao with dwarf coconut resulted in a higher net income compared to monoculture,

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due to the lower production costs and higher prices of both crops (Garcia et al., 2011). However, in some cases, the intercropping of cacao with coconut may result in lower net income compared to monoculture, due to the lower prices of coconut products and the higher competition for resources between the two crops (Kenny et al., 2006).

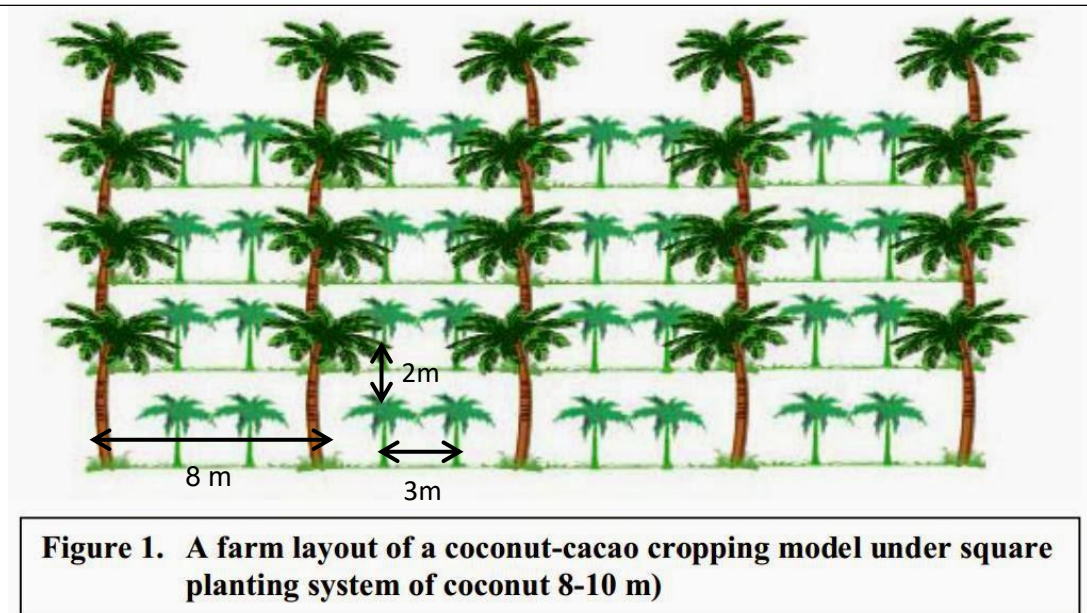
(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The study will be conducted for two factor experiment for RCBD. For factor A composed of different cacao varieties such as: A1 - K1, A2 - UF 18, A3 - BR 25, A4- Criollo, A5 - K2, and A6 - UIF1. Factor B, comprised the different Coconut varieties such as: [B1 - TAC](#), [B2 - ARO](#), [B3 - MRD](#)

BLOCK I	BLOCK II	BLOCK III
V1	V6	V2
V6	V5	V4
V5	V1	V5
V3	V3	V1
V4	V2	V6
V2	V4	V3

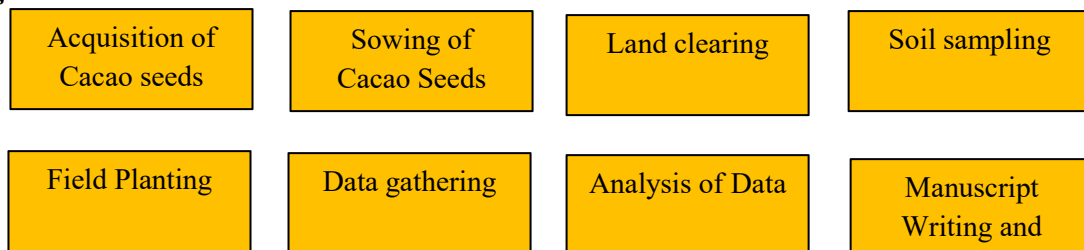
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Adapted from DA -PCA R&D, Coconut Cropping Guide No. 7

Experimental Site. The experiment will be conducted at Coconut Breeding Trial Unit (CBTU) of Capiz State University, Burias Campus, Burias, Mambusao, Capiz. Three coconut variety will be identify and will utilized in the study.

Activities



Data to be gathered:

Plant height (m). The plant height of cacao will be measure from the base of the plant to the tip of longest leaves.

Fan branches. This will be gather by counting the number of fan branches per cacao tree.

Stem girth (cm). Stem girth will be taken by measuring the stem diameter using a measuring tape every two weeks.

Number of flowers/cushion. This will be gather by counting the number of flowers per flower cushion.

Number of flower cushions/tree. This will be gather by counting the number of flower cushions per tree.

Number of pods harvested/tree. This will be gather by counting the number of pods harvested per tree.

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Pod weight (g). The pod weight will be gather by weighing the harvested pod per tree.

Number of beans/pod. This will be gather by counting the number beans per pods.

Bean size (cm). Bean size will be determine by measuring the diameter of beans using a caliper.

Return on Investment. This will be computed using the formula below:

$$\text{ROI} = \frac{\text{Net Income}}{\text{Total Expenses}} \times 100$$

Growth and yield parameters will be gathered and Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR). The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged double factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.

Recommended Practices in a Coconut -Cacao Cropping System

CACAO

1. In a small farm or a plantation, at least five recommended clonal materials to be grown at the same period, since cacao trees are normally incompatible in terms of flowering and pollination;
2. Under monoculture, a 3 m x 3 m triangular spacing (1,241 plants/ha) is desirable to be grown when intercrop with coconuts (coconut-cacao long term cropping system);
3. Recommended hybrid varieties are Criollo, Trinitario and Forastero. Commonly propagated by seeds grown in the nursery with regular watering as practical; seedlings are ready for transplanting in 6-8 months with 4-6 pairs of green true leaves are present; the use of budded clonal seedlings offers a true-to-type seedlings which achieve earlier higher yields;
4. In the initial years of crop establishment, shade of other tree crops (“nurse crops”) like coconut, *Gliricidia*, *Leucaeria* or *Erythrina* and other comparative crops is desirable; under coconut cacao monocraopping system, shade trees are later pruned;
5. Proper and timely pruning is required: 1) to train, shape and achieve the economical tree height; 2)to have adequate air circulation and sunlight penetration within the crop; 3) to minimize incidences of pest and diseases; and
4) to produce higher and quality yields;
6. Adequate fertilization is an important component of the integrated crop management of the cacao or cocoa tree crop.
7. The major insect pests of the crop are cacao pod borer (causes uneven ripening and unfilled beans), *Helopetis* (attcks immature pods) and *Apogonia* (leaf eaters) that are active at nighttime. While, the most important diseases are : black pod disease; red root disease (causing wilting and yellowing of

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leaves and eventual death of trees); white rot disease (causing wilting of leaves, leaving white mycelium and vascular streak disease (*Oncosbasidium theobromae*), characterize by brownish horizontal streak of infected twigs and causes the eventual death of shoots (Ministry of Agriculture of Malaysia).

8. Harvesting – is done when pods turns yellowish or reddish orange in color. Pick cacao fruits only when they are fully mature (147 days) Harvesting should be done regularly to avoid overripe pods in the trees. Usually, the cacao crop has peakharvest and off-season periods. The peak-harvest period is the result of flowering during the rainy season. Selective harvesting is done by hand using knives or machetes. Puling the pods from the tree damages the flower cushions and destroys the bark.

9. The following are some of the very important considerations in the postharvest handling of cacao beans: 1) avoid injuring the beans during pod-splitting (pods cut open to extract the beans); 2) wet cocoa beans should undergo fermentation for 7 days in order to kill the seeds and enhance the chocolate flavor; 3) dry under the sun or by a mechanical dryer; 4) dried beans are kept in gunny sacks and stack on raised platforms. Avoid damp conditions to control fungus attack which lower the bean quality. Dried fermented cacao beans (DFCB) are best graded before marketing, following the acceptable or standard grading system in the country. The criteria for grading are the number of beans per 100 g, % fungus infection, % pest incidence and % slaty beans (“flat” beans).

Fertilization

For Cacao

After Field Planting (month)	N	P ₂ O ₅	K ₂ O	MgO	14-14-14b
1	6.4	6.4	6.4	-	45
4	8.5	8.5	8.5	-	60
8	8.5	8.5	8.5	-	60
12	12.8	12.8	12.8	-	92
18	17.0	17.0	17.0	-	121
24	27.3	27.3	27.3	4.5	192 ^c
Total	80.5	80.5	80.5	4.5	570

^aPCAARRD 1989

^bEstimated from PCAARRD’s 1989 study

COCONUT

With the coconut trees are already established and already at bearing stage, the main farming practices are fertilization, underbrushing-weeding, mulching of the main rootzone of coconut (also considered the

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fertilizing zone at trunk base of trees), and harvesting. Post-harvest and primary processing practices (seasoning of partially immature nuts for 7–10 days, dehushing and copra processing) are common in small to medium scale farms. If sold to coconut desiccating plants, dehushed nuts are immediately marketed. Coconut husks await decortication/defibering, while coconut shells are converted to charcoal and sold to activated carbon processors. A separate fertilization for the stands of coconut and the cacao crop is recommended. There are two average inorganic/mineral fertilizer recommendations for coconut: 1) using the combination of single fertilizers (ammonium sulfate plus common salt (for potassium-rich soils) or potassium chloride (0-0-60) for soils deficient in K; and

2) using ready-to-apply multinutrient fertilizers as the 14-5-20-0.02 (B), now commercially available like COCOGROW (ATLAS Brand) in 25 kg. capacity bags.

These two fertilizer recommendations are compatible with the application of appropriate organic fertilizers (compost, cocopeat, commercial organic fertilizers). If capital resources to purchase organic fertilizers is available, any of these organic fertilizers (total N, P and K of at least 5%) may be applied together with the mineral fertilizers (options 1 and 2) indicated below at the rate of 3-4 times of the periodic rates indicated. Organic fertilizers should be applied about a month ahead of the application of the inorganic/mineral fertilizers. Organic fertilizers serve best as soil conditioners and fertilizer supplements to the coconut-cacao cropping system.

Fertilization

For Coconut

6 months from FP	200 g AS + 200 g NaCl or 200 g KCl
1 year	500 g AS + 450 g NaCl or 600 g KCl
2 years	750 g AS + 750 g NaCl or 900 g KCl
3 years	1.0 kg AS + 1.25 kg NaCl or 1.5 kg KCl
4 years	1.25 kg AS + 1.35 kg NaCl or 1.70 kg KCl
5 years and onwards	1.50 kg AS + 1.70 kg NaCl or 2.00 kg KCl

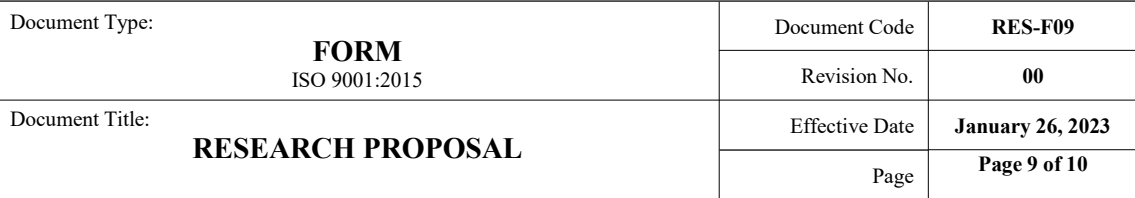
AS – Ammonium sulfate (21-0-0)

NaCl – Sodium chloride (common salt)

KCl – Potassium chloride (0-0-60)

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1												Year 2
	Jan	Feb	Mar	Apr	May	June	Jul	Aug	Sept	Oct	Nov	Dec	

[illegible]

1. Intercropping model for cacao-coconut production

1. Coconut farmers
2. Cacao farmers

Name	Role (Program,Project, Study Leader)	Field of Specialization
Dr. Salvacion J. Legaspi	Study Leader	Plant Physiology

Program/Project Members (in hours):

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1 Research assistant @ 504/day - for 3 Years	Php399,168.00

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	B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
-	Supplies and materials		250,000.00
-	Transportation allowance		20,000.00
-	Meal allowance during data gathering		5,000.00
-	Professional fees (laboratory analysis and other related fees)		150,000.00
-	Miscellaneous (for incidental expenses)		20,000.00
-	Printing		10,000.00
	C. CAPITAL OUTLAY		50,000.00
	Total		904,168.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from)

Adam, R. P., Panggeso, J., & Suardi, M. (2017, August). Analysis of cacao and coconut intercrop farming on production centers in Central Sulawesi Province. In *International Conference on Science and Technology (ICOSAT 2017)-Promoting Sustainable Agriculture, Food Security, Energy, and Environment Through Science and Technology for Development* (pp. 88-91). Atlantis Press

FAOSTAT (2021). Retrieved from <https://www.fao.org/faostat/en/#data/QC>

Garcia, J. R., P. M. Panganiban, and M. A. Bautista. 2011. Cacao-coconut intercropping: A viable alternative for smallholder farmers in the Philippines. *Journal of Agriculture and Food Security* 9(2): 168-177.

Kamanga, M. N., M. K. Mwale, and M. J. Kakulubi (2019). "Coconut and cacao intercropping: A review." *Journal of Agricultural Science and Technology*, 13(1), 1-12.

Kenny, B. J., J. E. R. D. M. T. T. R. M., and R. C. P. C. 2006. Cacao-coconut intercropping: A review. *Journal of Agriculture and Food Security* 4(1): 1-11.

Sutisna, M., A. K. S. S. R., and M. R. K. 2017. Cacao-coconut intercropping: A review. *Journal of Agriculture and Food Security* 15(1): 1-13

Sukardjo, R. S., I. S. Sihotang, and S. K. Sihotang (2018). "Coconut-cacao intercropping: A review." *Journal of Agricultural Science and Technology*, 12(1), 1-11

(21) Capsule Curriculum Vitae (One-page brief C.V. only)